

HAMZA KHAN

Software Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer who builds and ships production software end to end — data model, backend, frontend, packaging and deployment. Delivered Django, React/TypeScript and desktop applications for clients in real estate, industrial trading and education, with two sites live in production and a Windows application installed and running on site. Independently designed and released a local-first desktop BI product on Tauri, FastAPI and DuckDB behind a full CI pipeline. B.E. in Computer Science (Data Science), CGPI 8.19.

SKILLS & TECH STACK

Languages: Python, JavaScript, TypeScript, SQL, Rust (Tauri shell), HTML5, CSS3

Backend: Django 5, FastAPI, Flask, Django Channels, WebSockets, Daphne (ASGI), REST APIs, role-based access control, authentication and authorisation

Frontend: React 19, Next.js, Vite, Tailwind CSS, Radix UI, Ant Design, Framer Motion, React Router, Django templates, responsive UI, i18n, SEO

Databases: PostgreSQL, SQLite, DuckDB, Elasticsearch, SQLAlchemy, schema design, query optimisation (CTEs, window functions)

Data & Analytics: pandas, scikit-learn, Prophet, SHAP, mlxtend, lifetimes, TextBlob, openpyxl, ETL pipelines, RFM segmentation, forecasting, Power BI, ECharts, Chart.js

Desktop & Packaging: Electron, Tauri, PyInstaller, NSIS installer, offline-first and local-data architecture

Integrations: Google Drive API, Gmail API, Ollama (local LLM), pdfkit, wkhtmltopdf, ReportLab, APScheduler, GST invoicing, WhatsApp handoff

Testing & Tooling: pytest, Vitest, Ruff, ESLint, tsc --noEmit, CI pipelines, Git, GitHub, Docker

PROFESSIONAL EXPERIENCE

Software Engineer (Freelance) — Self-Employed

2024 – Present

Mumbai, Maharashtra, India

- Sole engineer on eight delivered projects for paying clients across real estate, industrial trading, manufacturing and education — owning requirements, data modelling, build, deployment and handover on every one.
- Shipped to production on three platforms: server-rendered Django applications, React and Next.js websites, and packaged Windows desktop applications installed on client machines.
- Designed content-as-data architectures so non-developer client staff can update live sites without code changes, reducing repeat maintenance work across multi-edition projects.
- Worked directly with non-technical business owners, translating operational processes — billing, tenancy, lead handling, conference administration — into working software and documented handovers.

SELECTED PROJECTS & PRODUCTS

Novem — Local-First Desktop BI Platform

Independent Product

Tauri (Rust), React 19, TypeScript, FastAPI, DuckDB, SQLite, scikit-learn, Prophet, SHAP, Ollama

- Built a desktop business-intelligence application for small e-commerce operators: a Tauri (Rust) shell wrapping a React 19 + TypeScript client, communicating with a local Python FastAPI engine of 23 routers and roughly 170 endpoints.
- Implemented demand forecasting with Prophet, RFM segmentation and churn scoring with scikit-learn, customer lifetime value via BG/NBD and Gamma-Gamma, market-basket rules with mlxtend, and SHAP explanations so each recommendation can justify itself.
- Architected all storage locally — DuckDB for analytical tables scoped by store, SQLite for settings, lineage and encrypted credentials — with a local Ollama LLM, so no customer data leaves the machine; customer emails are SHA-256 hashed on import.
- Designed graceful capability degradation: forecasting falls back to linear-trend extrapolation, sentiment from transformer to TextBlob to keyword matching, and the assistant to rule-based answers when optional components are absent.
- Packaged release builds by freezing the Python engine with PyInstaller inside an NSIS installer, gated by CI on every push — Ruff, pytest across Python 3.11 and 3.12, tsc --noEmit, ESLint at zero warnings, Vitest and a production build.

InitCore CRM — Outbound Call-Centre Operations Platform

Independent Product

Django 5, Python 3.12, PostgreSQL, Django Channels, WebSockets, Daphne, pdfkit, pandas, Chart.js

- Designed and built a 17-model Django platform covering the full desk workflow — lead import, dispositions, payments, GST invoicing, attendance and a live break monitor — in a single deployment.
- Implemented role-scoped queriesets so one view set serves superuser, team leader and agent, with data scope, visible pages and verification rights all derived from the user role.
- Built a CSV/XLSX import pipeline with a column-mapping step before any write and uniqueness enforced on contact numbers, eliminating duplicate records on repeated imports.
- Automated GST invoice generation on payment verification — amount converted to words in Indian numbering, template rendered and a PDF produced through pdfkit and wkhtmltopdf.
- Delivered real-time break tracking over Django Channels and WebSockets, with supervisors monitoring agent status live, and wrote up known deployment constraints for the client.

InitCore Real Estate CRM — Multi-Role Property Management Platform

Independent Product

Django, Python, PostgreSQL, server-rendered templates

- Built a co-living property management platform serving five distinct role-scoped portals — super user, property manager, maintenance supervisor, landlord and tenant — from one shared data model and install.
- Modelled inventory down to individual bed spaces, each carrying type, occupancy, rent and availability, with occupancy and collection rates computed against bed-space records rather than whole units.
- Implemented the tenancy lifecycle end to end: contract dates, rent, deposit, room assignment and financial breakdown, with expiring contracts surfaced on both administrator and tenant views.
- Delivered invitation-based onboarding with expiry tracking, a five-stage lead pipeline with ownership and activity trails, and ticket-driven maintenance with priority and assignment.

Key2YourHome — Property Discovery Platform (key2yourhome.net)

Client Project

React, TypeScript, Tailwind CSS, responsive UI, SEO, cookie consent

- Designed, built and deployed a live production property portal for a MahaRERA-registered channel partner covering Mumbai, Thane, Navi Mumbai and Pune.
- Built searchable indexes for projects, resale, rentals and land, filterable by city, type, configuration and budget, with project pages carrying specifications, connectivity data, unit plans and FAQs.
- Implemented a sticky enquiry rail with WhatsApp handoff, site-visit booking, brochure download and a callback form pre-filled with the project context, so each lead arrives attributed.
- Delivered regulatory compliance as a product feature — MahaRERA registration surfaced on the hero and in a QR card linking to the state portal, with granular cookie consent separating essential, analytics and marketing.

S.K Trading — Offline GST Billing Desktop Application

Client Project

Electron, SQLite, JavaScript, PDF generation, Google Drive API, Gmail API

- Built and installed a Windows desktop application replacing a spreadsheet-and-Word document routine for an industrial valve supplier, covering quotations, proforma invoices, tax invoices and delivery challans.
- Implemented a split editor with a live A4 print preview so the approved document is exactly what prints, with line items carrying size, HSN code, quantity, unit, rate and GST percentage.
- Automated tax computation — splitting CGST/SGST or IGST from place of supply and writing the grand total in words on the document.
- Derived customer and product records from issued documents rather than manual entry, surfacing invoice counts, total billed, outstanding balances and the last rate quoted per line item.
- Built reporting by customer, product and state with CSV and segmented contact exports, running fully offline on local SQLite with opt-in Google Drive backup and Gmail sending.

EDUCATION

B.E., Computer Science (Data Science) — CGPI 8.19 / 73.73%

2023 – 2026

Saraswati College of Engineering (Autonomous, affiliated to University of Mumbai), Navi Mumbai

Diploma in Computer Engineering — 72.63%, First Class

2021 – 2023

Maharashtra State Board of Technical Education (MSBTE)

HSC (Science) — 87.50% | SSC — 81.20%, Distinction

2021 / 2019

Maharashtra State Board of Secondary and Higher Secondary Education

CERTIFICATIONS & ACHIEVEMENTS

- **IELTS Academic — Overall Band 6.5, CEFR Level B2 (June 2026).** Speaking 7.0, Listening 6.5, Reading 6.5, Writing 6.0.
- **Smart India Hackathon 2025 — Grand Finale Participant, Software Edition (December 2025).** National hackathon by the Ministry of Education, Government of India.
- **SCOE Avishkar 2025 Project Competition — 2nd Place (April 2025).** Saraswati College of Engineering, Navi Mumbai.